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Case Study: Cyclistic

Scenario

You are a junior data analyst working on the marketing analyst team at Cyclistic, a bike-share company in Chicago. The director of marketing believes the company's future success depends on maximizing the number of annual memberships. Therefore, your team wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, your team will design a new marketing strategy to convert casual riders into annual members. But first, Cyclistic executives must approve your recommendations, so they must be backed up with compelling data insights and professional data visualizations.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members will be key to future growth. Rather than creating a marketing campaign that targets all-new customers, Moreno believes there is a solid opportunity to convert casual riders into members.

Ask

Three questions will guide the future marketing program:

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

Moreno has assigned you the first question to answer: How do annual members and casual riders use Cyclistic bikes differently?

You will produce a report with the following deliverables:

1. A clear statement of the business task
2. A description of all data sources used
3. Documentation of any cleaning or manipulation of data
4. A summary of your analysis
5. Supporting visualizations and key findings

6. Your top three recommendations based on your analysis Use the following Case Study Roadmap as a guide.

Case Study Roadmap – Ask

Guiding questions

- What is the problem you are trying to solve?
 - Cyclistic wants to convert more casual riders into annual members.
- How can your insights drive business decisions?
 - By identifying when, where, how often, and how long casual riders ride compared with members, marketing can target the right riders with the right message.

Key tasks

- Identify the business task
 - Analyze how annual members and casual riders use Cyclistic bikes differently
- Consider key stakeholders
 - They are the marketing director, executive team, and marketing analytics team.

Deliverable

- A clear statement of the business task
 - Identify key differences between casual riders and annual members using 2025 Divvy trip data, then recommend marketing strategies to convert casual riders into annual Cyclistic members.

Prepare

Use Cyclistic's historical trip data to analyze and identify trends. [Download the previous 12 months of Cyclistic trip data here](#). (Note: The datasets have a different name because Cyclistic is a fictional company. For the purposes of this case study, the datasets are appropriate and will enable you to answer the business questions. The data has been made available by Motivate International Inc. under this [license](#).) This is public data that you can use to explore how different customer types are using Cyclistic bikes. But note that data-privacy issues prohibit you from using riders' personally identifiable information. This means that you won't be able to connect pass purchases to credit card numbers to determine if casual riders live in the Cyclistic service area or if they have purchased multiple single passes.

Case Study Roadmap – Prepare

Guiding questions

- Where is your data located?
 - The data is stored in Amazon AWS. It comes from the official Divvy historical trip database.

I have downloaded every monthly file of 2025 locally to my computer.

● How is the data organized?

-There are 12 tables corresponding to each month of the year. Each table is a .csv file with information about the ride_id, start and end date and time, start and end location, what type of ride was taken, and whether the ride was taken by a member or casual user. Each row represents one completed bike ride.

● Are there issues with bias or credibility in this data? Does your data ROCCC?

-There is no inherent bias. This is credible data because it is first-party data provided by Motivate International Inc., a subsidiary of Lyft.

Reliable: Moderately high. This is unbiased data because it was collected through automatic observations rather than survey responses, but some columns have null values. Original: High. Motivate International Inc utilizes Amazon Web Services to host portions of its infrastructure.

Comprehensive: Medium. The data source contains some critical information needed to answer the question and avoids seasonal bias. However, data-privacy issues prohibit me from using riders' personally identifiable information. This means that I won't be able to connect pass purchases to credit card numbers to determine if casual riders live in the Cyclistic service area or if they have purchased multiple single passes. Demographics, income, age, and gender are unavailable.

Current: High. Yes, the data source is updated each month. For this case study, I will be using the most recent complete calendar year available, 2025.

Cited: High. This data was created by Motivate International Inc. The data comes directly from a credible organization and was refreshed recently. It can be found online.

● How are you addressing licensing, privacy, security, and accessibility?

-The data-privacy issues prohibit me from using riders' personally identifiable information. Instead, each ride is given a unique ride ID. Since I cannot glean demographic information from the dataset, I will only be able to make suggestions relying strictly on riding behavior. The Divvy dataset is publicly available for education and analytical purposes under its published data license. Security-wise, the data will remain inside Google Cloud/BigQuery, and no modification will be made to the original raw files.

● How did you verify the data's integrity?

-First, I created a new project in BigQuery. Then, I created a storage bucket in which I uploaded all 12 monthly trip data .csv files. Then, I created a dataset by loading the files in the storage bucket as a native table. Then I verified that the table loaded correctly by ensuring it had the correct columns. Then I verified that the columns had the correct data type.

● How does it help you answer your question?

-Because each row represents a single ride and includes the rider type, this dataset can

help me compare annual members and casual riders by answering questions such as: what stations different member groups usually start and end, how long their trips are, and what days of week they usually ride, and what months have the most bikeshare usage. From those fields, I will derive additional metrics such as ride length, day of week, month, hour of day, season, and weekend vs. weekday.

- Are there any problems with the data?

-Yes. Although the dataset is high quality, there are null values, duplicates, and invalid ride durations. This dataset also lacks demographic information.

Key tasks

- Download data and store it appropriately.

- Identify how it's organized.

- Sort and filter the data.

- Determine the credibility of the data.

Deliverable

- A description of all data sources used

-This dataset contains anonymized records of every completed bike-share trip taken through the Divvy bike-sharing system during 2025 which can be found here [Index of bucket "divvy-tripdata"](#). Each record represents one ride and includes information such as ride ID, rider type, bike type, timestamps, station information, and geographic coordinates. The dataset is maintained by Divvy and made publicly available for educational and analytical purposes.

Process

Case Study Roadmap - Process

Guiding questions

- What tools are you choosing and why?

-Big Query SQL. The complete 2025 Divvy dataset contains millions of records. BigQuery is designed to efficiently process datasets of this size without row or memory limitations of spreadsheets.

- Have you ensured your data's integrity?

-Yes.

- What steps have you taken to ensure that your data is clean?

-I preserved the raw table, then created a clean working table. I checked for duplicate ride_id values, removed duplicate rides, checked missing values, and removed rows missing critical fields. I also calculated ride length and removed rides that lasted under 1 minute or over 24 hours. I also derived some date fields for analysis, then validated the

final cleaned table.

● How can you verify that your data is clean and ready to analyze?

-I verified that duplicates were removed by checking the cleaned table's total rows. I also verified that there were no nulls in critical fields. I also validated that ride lengths and dates were within reasonable limits. Afterwards, I ensured the data was ready to analyze by making sure it aligned with the business objective. I also ensured the sample size was at least 30 so that the results would be statistically significant. I used the data-cleaning verification checklist to validate the cleaned data.

● Have you documented your cleaning process so you can review and share those results?

-Yes, the cleaning process and changelog are in the file called `cyclistic-data-cleaning-changelog.pdf`

Key tasks

● Check the data for errors.

● Choose your tools.

● Transform the data so you can work with it effectively.

● Document the cleaning process.

Deliverable

● Documentation of any cleaning or manipulation of data

Analyze

Case Study Roadmap - Analyze

Guiding questions

● How should you organize your data to perform analysis on it?

-I used the cleaned table as the base table and created smaller BigQuery Views for Tableau. This keeps the project organized, reproducible, and easier to explain.

● Has your data been properly formatted?

-Yes. The process phase created clean fields for analysis.

● What surprises did you discover in the data?

-Casual riders take longer average rides than members. Members account for more total rides. Casual riders are more concentrated on weekends. Members show clearer peaks in commute hours. Casual ridership rises more sharply in summer. Casual-heavy stations may cluster near parks, lakefronts, museums, or tourist zones.

● What trends or relationships did you find in the data?

-Members ride more frequently. Casual riders ride longer. Members use bikes more during weekdays, whereas casual riders are more weekend oriented. Members show commuting behavior. Casual riders show leisure behavior.

● How will these insights help answer your business questions?

-These insights show how casual riders behave differently from members, which helps Cyclistic design targeted campaigns. For example: if casual riders ride mostly on weekends, campaigns should launch before weekend trips. If casual riders cluster near tourist/leisure stations, station-based ads should target those areas. If casual riders take longer rides, marketing can emphasize savings from membership for frequent long-distance leisure users. If casual ridership peaks in summer, membership campaigns should begin in spring before demand peaks.

Key tasks

- Aggregate your data so it's useful and accessible.
- Organize and format your data.
- Perform calculations.
- Identify trends and relationships.

Deliverable

● A summary of your analysis

-I used BigQuery SQL to compare casual riders and annual members by total ride count, average ride length, median ride length, month, day of week, weekday/weekend, hour of day, bike type, station usage, and seasonal patterns.

Share

Case Study Roadmap - Share

Guiding questions

● Were you able to answer the question of how annual members and casual riders use Cyclistic bikes differently?

-Yes

● What story does your data tell?

-Members use Cyclistic more like routine transportation, while casual riders use it more for leisure, weekends, warmer months, and destination-based trips.

● How do your findings relate to your original question?

-The visuals compare annual members and casual riders by ride volume, ride length, month, day of week, hour of day, bike type, and station usage. Together these show how each rider group uses Cyclistic differently by showing frequency, duration, seasonality, weekly behavior, daily timing, product usage, and location behavior.

● Who is your audience? What is the best way to communicate with them?

-My audience is Cyclistic's marketing and executive team. The best way to communicate with them is by using a short presentation with a Tableau dashboard with 3-5 clear insights

and 3 marketing recommendations.

- Can data visualization help you share your findings?

-Yes.

- Is your presentation accessible to your audience?

-Yes.

Key tasks

- Determine the best way to share your findings.

- Create effective data visualizations.

- Present your findings.

- Ensure your work is accessible.

Deliverable

- Supporting visualizations and key findings

-These can be found in the file called cyclistic-presentation.pdf.

Act

Case Study Roadmap - Act

Guiding questions

- What is your final conclusion based on your analysis?

-Because members ride more during the weekday and peak during commute hours, and casual riders ride more during weekends or later in the day, take longer trips on average, and are concentrated around leisure or tourist locations, these patterns suggest that members primarily use Cyclistic as recurring transportation, and casual riders are more likely to use the service for recreational trips.

- How could your team and business apply your insights?

-Since casual ridership peaks during spring and summer, Cyclistic should launch membership campaigns shortly before the strongest casual ridership periods. Since casual riders take longer average and median rides than members, Cyclistic should message the annual membership as a better value for casual riders who already take longer or repeated trips. Since my findings found casual-heavy stations, Cyclistic should focus ads around tourist, leisure, lakefront, park, and entertainment heavy stations.

- What next steps would you or your stakeholders take based on your findings?

-Cyclistic should begin with a controlled pilot campaign rather than launching every recommendation citywide at once. They should also do A/B tests for savings vs convenience, seasonal vs year-round messaging, post-ride vs pre-ride promotion, general vs location specific messaging. They should also assess more KPI's like incremental conversion instead of relying only on membership growth to measure the how the

campaign is going.

- Is there additional data you could use to expand on your findings?

-I could collect and analyze customer-level usage, pricing, and marketing attribution data to optimize the marketing campaign. Other useful data may include weather, events and tourism, and station capacity data.

Key tasks

- Create your portfolio.

- Add your case study.

- Practice presenting your case study to a friend or family member.

Deliverable

- Your top three recommendations based on your analysis

-Target casual riders in spring before summer usage increases. Message the annual membership as a better value for casual riders who already take longer or repeated trips. Focus ads around tourist, leisure, lakefront, park, and entertainment heavy stations.